

A Finite-Dimensional Technical Note on Global Smoothness of Gaussian Associative-Memory Cross-Entropy under Frobenius and Operator-Norm Geometries

Abstract

We give a self-contained finite-dimensional proof of global smoothness bounds for a Gaussian associative-memory softmax loss. The proof is organized around one explicit good event, and every failure probability is stated in its original exponential-tail form; no confidence parameter is introduced. For Zipf weights

$$p_i = \frac{i^{-\alpha}}{H_{N,\alpha}}, \quad H_{n,\alpha} = \sum_{k=1}^n k^{-\alpha}, \quad \alpha \geq 0,$$

we prove, simultaneously with an explicit probability, that

$$\beta_F \asymp \frac{1}{d} + p_1, \quad \beta_{\text{op}} \gtrsim \frac{1}{d} + P_{\min\{d,N\}},$$

and we establish an explicit finite-dimensional upper bound for β_{op} . In the regime $N = d^{\Theta(1)}$, the operator-norm bound matches the lower bound up to polylogarithmic factors.

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1 Setup, definitions, and main theorem

1.1 Random model and loss

For an integer $n \geq 1$, write $[n] := \{1, \dots, n\}$. Fix integers $d, N \geq 2$. Let

$$u_1, \dots, u_N, v_1, \dots, v_N$$

be mutually independent random vectors with

$$u_j \sim \mathcal{N}\left(0, \frac{1}{d}I_d\right), \quad v_i \sim \mathcal{N}\left(0, \frac{1}{d}I_d\right).$$

The vectors u_j are the output embeddings and the vectors v_i are the input embeddings. Put

$$U := [u_1, \dots, u_N] \in \mathbb{R}^{d \times N}.$$

Fix a Zipf exponent $\alpha \geq 0$, and define

$$H_{n,\alpha} := \sum_{k=1}^n k^{-\alpha}, \quad p_i := \frac{i^{-\alpha}}{H_{N,\alpha}}, \quad i \in [N].$$

Thus $p_1 \geq \dots \geq p_N > 0$ and $\sum_{i=1}^N p_i = 1$. For $s \in [N]$, define the head mass

$$P_s := \sum_{i=1}^s p_i = \frac{H_{s,\alpha}}{H_{N,\alpha}}.$$

For $W \in \mathbb{R}^{d \times d}$, define the logits and softmax probabilities

$$z_{ij}(W) := u_j^\top W v_i, \quad q_{ij}(W) := \frac{e^{z_{ij}(W)}}{\sum_{k=1}^N e^{z_{ik}(W)}}.$$

The loss associated with input i and the population loss are

$$\ell_i(W) := \log \left(\sum_{j=1}^N e^{u_j^\top W v_i} \right) - u_i^\top W v_i, \quad L(W) := \sum_{i=1}^N p_i \ell_i(W).$$

The target term $-u_i^\top W v_i$ is linear in W , and therefore it does not appear in the Hessian.

We use the Frobenius inner product

$$\langle A, B \rangle_F := \text{Tr}(A^\top B).$$

The Frobenius and operator norms are denoted by $\|\cdot\|_F$ and $\|\cdot\|_{\text{op}}$, respectively. Define the global quadratic smoothness constants

$$\beta_F := \sup_{W \in \mathbb{R}^{d \times d}} \sup_{H \neq 0} \frac{\nabla^2 L(W)[H, H]}{\|H\|_F^2},$$

$$\beta_{\text{op}} := \sup_{W \in \mathbb{R}^{d \times d}} \sup_{H \neq 0} \frac{\nabla^2 L(W)[H, H]}{\|H\|_{\text{op}}^2}.$$

Section 2 proves that these are exactly the global Lipschitz constants of the gradient from the corresponding primal norm to its dual norm.

1.2 Parameters and explicit failure probability

Set

$$m := \min\{d, N\}, \quad K := \left\lfloor \frac{m}{256} \right\rfloor.$$

When $N > d$, define

$$J := \left\lceil \log_2 \frac{N}{d} \right\rceil, \quad \Xi_{\alpha, J} := \sum_{r=0}^{J-1} 2^{-\alpha r}.$$

When $N \leq d$, set $J := 0$ and $\Xi_{\alpha, J} := 0$. In all cases, set

$$L_N := \log(2N^2), \quad T_d := d^2 \log(3d) + d.$$

The selector construction below has failure probability

$$\varepsilon_{\text{sel}}(d, N, K) := 4Ke^{-d/32} + 2Ke^{-d/48} + K(N-2)e^{-d/64}.$$

Define the total explicit failure bound

$$\begin{aligned} \varepsilon_{\text{tot}}(d, N) &:= 2Ne^{-d/2} + e^{-d} + e^{-d/16} + \varepsilon_{\text{sel}}(d, N, K) \\ &\quad + 4 \cdot 17^K e^{-d/32} + Je^{-d}. \end{aligned} \tag{1}$$

No independence between the events contributing to (1) is asserted or needed; the final estimate follows from a union bound and one conditional union bound for the dyadic blocks.

Theorem 1.1 (Finite-dimensional global smoothness). *Assume*

$$\alpha \geq 0, \quad m = \min\{d, N\} \geq 512.$$

Then, with probability at least

$$\boxed{1 - \varepsilon_{\text{tot}}(d, N)},$$

all of the following conclusions hold simultaneously.

Frobenius geometry.

$$\boxed{\frac{1}{64} \left(\frac{1}{d} + p_1 \right) \leq \beta_F \leq 80 \left(\frac{1}{d} + p_1 \right)}.$$

Equivalently, because $p_1 = H_{N, \alpha}^{-1}$,

$$\frac{1}{64} \left(\frac{1}{d} + \frac{1}{H_{N, \alpha}} \right) \leq \beta_F \leq 80 \left(\frac{1}{d} + \frac{1}{H_{N, \alpha}} \right).$$

Operator geometry: lower bound.

$$\boxed{\beta_{\text{op}} \geq \frac{1}{294912} \left(\frac{1}{d} + P_m \right)}.$$

Operator geometry: upper bound. *In every case,*

$$\boxed{\beta_{\text{op}} \leq 16.}$$

If $N > d$, then the sharper estimate

$$\boxed{\beta_{\text{op}} \leq \min \left\{ 16, 16P_d + \frac{2^{\alpha+4}(L_N + 10)}{d} + \frac{32T_d}{d^2} \Xi_{\alpha,J} P_d \right\}}$$

also holds. Here $N > d$ implies $m = d$, so $P_d = P_m$.

Remark 1.2 (When the probability bound is informative). *The theorem is a finite-dimensional statement and does not require a separate relation between d and N . If the displayed lower bound on probability is negative, it is simply vacuous. In the common regime $N = d^{O(1)}$, every term in (1) is $e^{-\Omega(d)}$, because $K \leq d/256$ and $J = O(\log d)$.*

2 Exact Hessian formula and norm geometry

For $i \in [N]$, write

$$q_i(W) := (q_{i1}(W), \dots, q_{iN}(W))^\top \in \mathbb{R}^N$$

and define

$$C_i(W) := U(\text{Diag}(q_i(W)) - q_i(W)q_i(W)^\top)U^\top.$$

Proposition 2.1 (Gradient and Hessian). *For every $W, A, B \in \mathbb{R}^{d \times d}$,*

$$\begin{aligned} \nabla \ell_i(W) &= \left(\sum_{j=1}^N q_{ij}(W)u_j - u_i \right) v_i^\top, \\ \langle B, \nabla^2 L(W)[A] \rangle_F &= \sum_{i=1}^N p_i(Bv_i)^\top C_i(W)(Av_i). \end{aligned}$$

Moreover, $C_i(W) \succeq 0$, and hence

$$\boxed{\nabla^2 L(W)[H, H] = \sum_{i=1}^N p_i(Hv_i)^\top C_i(W)(Hv_i) \geq 0.}$$

Proof. For a direction A ,

$$Dz_{ij}(W)[A] = u_j^\top Av_i.$$

Therefore

$$\begin{aligned} D\ell_i(W)[A] &= \sum_{j=1}^N q_{ij}(W)u_j^\top Av_i - u_i^\top Av_i \\ &= \left(\sum_{j=1}^N q_{ij}(W)u_j - u_i \right)^\top Av_i. \end{aligned}$$

Using the Frobenius inner product gives the displayed gradient formula.

Let

$$\mu_i(W) := \sum_{j=1}^N q_{ij}(W) u_j = U q_i(W).$$

The Jacobian of the softmax map is

$$D \text{softmax}(z) = \text{Diag}(q) - qq^\top.$$

Consequently,

$$Dq_i(W)[A] = (\text{Diag}(q_i(W)) - q_i(W)q_i(W)^\top)U^\top Av_i,$$

and hence

$$D\mu_i(W)[A] = C_i(W)Av_i.$$

Because the target term is linear,

$$\nabla^2 \ell_i(W)[A] = C_i(W)Av_i v_i^\top.$$

Summing over i and taking the Frobenius inner product with B yields

$$\langle B, \nabla^2 L(W)[A] \rangle_F = \sum_{i=1}^N p_i(Bv_i)^\top C_i(W)(Av_i).$$

Finally,

$$C_i(W) = \sum_{j=1}^N q_{ij}(W) u_j u_j^\top - \left(\sum_{j=1}^N q_{ij}(W) u_j \right) \left(\sum_{j=1}^N q_{ij}(W) u_j \right)^\top$$

is a covariance matrix, so it is positive semidefinite. $\square$

Proposition 2.2 (Quadratic characterization of smoothness). *Let $\|\cdot\|_X$ be any norm on $\mathbb{R}^{d \times d}$, and let $\|\cdot\|_{X,*}$ be its dual norm. For every fixed W ,*

$$\sup_{\|H\|_X \leq 1} \|\nabla^2 L(W)[H]\|_{X,*} = \sup_{\|H\|_X \leq 1} \nabla^2 L(W)[H, H].$$

Consequently,

$$\sup_W \sup_{\|H\|_X \leq 1} \nabla^2 L(W)[H, H]$$

is the smallest number β such that

$$\|\nabla L(W_1) - \nabla L(W_0)\|_{X,*} \leq \beta \|W_1 - W_0\|_X$$

for all $W_0, W_1 \in \mathbb{R}^{d \times d}$.

Proof. For fixed W , define the symmetric positive-semidefinite bilinear form

$$B_W(A, H) := \langle A, \nabla^2 L(W)[H] \rangle_F.$$

Positive semidefiniteness follows from Proposition 2.1. Applying nonnegativity to $B_W(A+tH, A+tH)$ as a quadratic polynomial in t gives

$$|B_W(A, H)|^2 \leq B_W(A, A)B_W(H, H).$$

Therefore

$$\begin{aligned} \sup_{\|H\|_X \leq 1} \|\nabla^2 L(W)[H]\|_{X,*} &= \sup_{\|A\|_X \leq 1, \|H\|_X \leq 1} |B_W(A, H)| \\ &\leq \sup_{\|G\|_X \leq 1} B_W(G, G). \end{aligned}$$

The reverse inequality follows by choosing $A = H = G$.

For the Lipschitz statement, use the fundamental theorem of calculus:

$$\nabla L(W_1) - \nabla L(W_0) = \int_0^1 \nabla^2 L(W_0 + t(W_1 - W_0))[W_1 - W_0] dt.$$

The preceding operator-norm identity gives the upper bound. Conversely, a global gradient Lipschitz bound implies the same Hessian bound by taking a directional derivative at any W . $\square$

3 Deterministic reductions

Define

$$R_u := \max_{j \in [N]} \|u_j\|^2, \quad M_v := \sum_{i=1}^N p_i v_i v_i^\top.$$

For $y \in \mathbb{R}^d$, define the output width

$$w_U(y) := \max_{j,k \in [N]} (u_j - u_k)^\top y = \max_j u_j^\top y - \min_k u_k^\top y.$$

Lemma 3.1 (Uniform covariance bound). *For every probability vector $q \in \mathbb{R}^N$, define*

$$C(q) := \sum_{j=1}^N q_j u_j u_j^\top - \left(\sum_{j=1}^N q_j u_j \right) \left(\sum_{j=1}^N q_j u_j \right)^\top.$$

Then

$$0 \preceq C(q) \preceq R_u I_d.$$

Proof. For every $x \in \mathbb{R}^d$,

$$\begin{aligned} x^\top C(q) x &= \sum_j q_j (x^\top u_j)^2 - \left(\sum_j q_j x^\top u_j \right)^2 \\ &\leq \sum_j q_j (x^\top u_j)^2 \leq R_u \|x\|^2. \end{aligned}$$

The lower bound follows because $C(q)$ is a covariance matrix. $\square$

Lemma 3.2 (Frobenius reduction). *For every realization of the embeddings,*

$$\boxed{\beta_F \leq R_u \|M_v\|_{\text{op}}}.$$

Proof. By Proposition 2.1 and Lemma 3.1,

$$\begin{aligned}\nabla^2 L(W)[H, H] &\leq R_u \sum_{i=1}^N p_i \|H v_i\|^2 \\ &= R_u \operatorname{Tr}(H^\top H M_v) \\ &\leq R_u \|M_v\|_{\text{op}} \|H\|_F^2.\end{aligned}$$

Take the supremum over W and nonzero H . □

Lemma 3.3 (Operator-width reduction). *For every realization,*

$$\boxed{\beta_{\text{op}} \leq \frac{1}{4} \sup_{\|H\|_{\text{op}} \leq 1} \sum_{i=1}^N p_i w_U(H v_i)^2.}$$

Also,

$$\boxed{\beta_{\text{op}} \leq R_u \operatorname{Tr}(M_v).}$$

Proof. Let a scalar random variable X be supported in $[a, b]$. Then

$$\operatorname{Var}(X) = \min_c \mathbb{E}(X - c)^2 \leq \mathbb{E} \left(X - \frac{a+b}{2} \right)^2 \leq \frac{(b-a)^2}{4}.$$

For fixed y , apply this to the random variable $u_j^\top y$ under $j \sim q_i(W)$. Its range length is $w_U(y)$, so

$$y^\top C_i(W) y \leq \frac{1}{4} w_U(y)^2.$$

Substitute $y = H v_i$ into Proposition 2.1 and then take the supremum over W and $\|H\|_{\text{op}} \leq 1$.

The second estimate follows from

$$\begin{aligned}\nabla^2 L(W)[H, H] &\leq R_u \sum_i p_i \|H v_i\|^2 \\ &\leq R_u \|H\|_{\text{op}}^2 \sum_i p_i \|v_i\|^2 = R_u \|H\|_{\text{op}}^2 \operatorname{Tr}(M_v).\end{aligned}$$

□

4 Probability tools

Lemma 4.1 (Weighted chi-square inequality). *Let $\xi_1, \dots, \xi_M$ be independent standard normal variables, and let $a_1, \dots, a_M \geq 0$. Define*

$$S := \sum_{r=1}^M a_r (\xi_r^2 - 1).$$

Then, for every $t > 0$,

$$\begin{aligned}\mathbb{P} \left(S \geq 2 \|a\|_2 \sqrt{t} + 2 \|a\|_\infty t \right) &\leq e^{-t}, \\ \mathbb{P} \left(S \leq -2 \|a\|_2 \sqrt{t} \right) &\leq e^{-t}.\end{aligned}$$

Proof. If $\|a\|_2 = 0$, then every $a_r = 0$, so $S = 0$ and both claims are immediate. Assume $\|a\|_2 > 0$.

For $0 < \lambda < (2\|a\|_\infty)^{-1}$,

$$\mathbb{E}e^{\lambda a_r(\xi_r^2 - 1)} = e^{-\lambda a_r}(1 - 2\lambda a_r)^{-1/2}.$$

For $0 \leq x < 1/2$,

$$-\frac{1}{2}\log(1 - 2x) - x \leq \frac{x^2}{1 - 2x}.$$

By independence,

$$\log \mathbb{E}e^{\lambda S} \leq \frac{\lambda^2 \|a\|_2^2}{1 - 2\lambda \|a\|_\infty}.$$

Chernoff's inequality with

$$\lambda = \frac{\sqrt{t}}{\|a\|_2 + 2\|a\|_\infty \sqrt{t}}$$

gives the upper tail.

For the lower tail,

$$\mathbb{E}e^{-\lambda a_r(\xi_r^2 - 1)} = e^{\lambda a_r}(1 + 2\lambda a_r)^{-1/2}.$$

Since

$$x - \frac{1}{2}\log(1 + 2x) \leq x^2 \quad (x \geq 0),$$

we have

$$\log \mathbb{E}e^{-\lambda S} \leq \lambda^2 \|a\|_2^2.$$

Chernoff's inequality with $\lambda = \sqrt{t}/\|a\|_2$ gives the lower tail. □

Lemma 4.2 (Covering nets). *The Euclidean unit sphere in $\mathbb{R}^q$ has an r -net in Euclidean norm of cardinality at most*

$$\left(1 + \frac{2}{r}\right)^q.$$

The operator-norm unit ball in $\mathbb{R}^{d \times d}$ has a $d^{-1/2}$ -net in operator norm of cardinality at most

$$(3d)^{d^2}.$$

Proof. A maximal r -separated subset of a normed set is an r -net. For the Euclidean sphere, the Euclidean balls of radius $r/2$ centered at the net points are disjoint and contained in the Euclidean ball of radius $1 + r/2$. Volume comparison gives the first claim.

For the second claim, take a maximal $d^{-1/2}$ -separated subset in operator norm of the operator unit ball. Since

$$\|A\|_F \leq \sqrt{d} \|A\|_{\text{op}},$$

the operator unit ball lies in the Frobenius ball of radius $\sqrt{d}$ in ambient dimension d^2 . Moreover, operator-norm separation implies Frobenius-norm separation. Euclidean volume comparison therefore gives

$$\left(1 + \frac{2\sqrt{d}}{d^{-1/2}}\right)^{d^2} = (1 + 2d)^{d^2} \leq (3d)^{d^2}.$$

□

Lemma 4.3 (Gaussian vector estimates). *Let g, h be independent $\mathcal{N}(0, I_d/d)$ vectors. Then*

$$\mathbb{P}\left(\|g\|^2 \notin [1/2, 3/2]\right) \leq 2e^{-d/32},$$

$$\mathbb{P}(\|g\| > 2) \leq e^{-d/2},$$

and

$$\mathbb{P}\left(\left|g^\top h\right| > \frac{1}{4}, \|g\|^2 \leq \frac{3}{2}\right) \leq 2e^{-d/48}.$$

Proof. Because $d\|g\|^2 \sim \chi_d^2$, Chernoff optimization gives

$$\mathbb{P}(\chi_d^2 \geq ad) \leq \exp\left[-\frac{d}{2}(a - 1 - \log a)\right] \quad (a > 1),$$

and

$$\mathbb{P}(\chi_d^2 \leq ad) \leq \exp\left[-\frac{d}{2}(a - 1 - \log a)\right] \quad (0 < a < 1).$$

For $a = 3/2$ and $a = 1/2$, the exponent is at least $d/32$. For $a = 4$, the exponent is at least $d/2$. This proves the first two claims.

Conditional on g ,

$$g^\top h \sim \mathcal{N}\left(0, \frac{\|g\|^2}{d}\right).$$

On the event $\|g\|^2 \leq 3/2$, the two-sided Gaussian tail bound gives

$$\mathbb{P}\left(\left|g^\top h\right| > \frac{1}{4} \mid g\right) \leq 2 \exp\left(-\frac{(1/4)^2}{2(3/2)/d}\right) = 2e^{-d/48}.$$

□

Lemma 4.4 (Rectangular Gaussian conditioning). *Let $G \in \mathbb{R}^{d \times K}$ have independent $\mathcal{N}(0, 1/d)$ entries, and assume $1 \leq K \leq d/256$. Then*

$$\mathbb{P}\left(\sigma_{\min}(G) < \frac{1}{2} \text{ or } \|G\|_{\text{op}} > \frac{3}{2}\right) \leq 2 \cdot 17^K e^{-d/32}.$$

Proof. Take a $1/8$ -net $\mathcal{N}$ of $\mathbb{S}^{K-1}$ with $|\mathcal{N}| \leq 17^K$. For every fixed $x \in \mathbb{S}^{K-1}$,

$$Gx \sim \mathcal{N}(0, I_d/d).$$

Lemma 4.3 and a union bound imply that, except on an event of probability at most $2 \cdot 17^K e^{-d/32}$,

$$\frac{1}{\sqrt{2}} \leq \|Gx_0\| \leq \sqrt{\frac{3}{2}} \quad \text{for every } x_0 \in \mathcal{N}.$$

Assume this event. For arbitrary $x \in \mathbb{S}^{K-1}$, choose $x_0 \in \mathcal{N}$ with $\|x - x_0\| \leq 1/8$. First,

$$\|G\|_{\text{op}} \leq \sqrt{\frac{3}{2}} + \frac{1}{8} \|G\|_{\text{op}},$$

so

$$\|G\|_{\text{op}} \leq \frac{8}{7} \sqrt{\frac{3}{2}} < \frac{3}{2}.$$

Second,

$$\|Gx\| \geq \frac{1}{\sqrt{2}} - \frac{1}{8} \|G\|_{\text{op}} \geq \frac{1}{\sqrt{2}} - \frac{1}{7} \sqrt{\frac{3}{2}} > \frac{1}{2}.$$

Taking the infimum over $x \in \mathbb{S}^{K-1}$ proves the lower singular-value bound. □

5 Frobenius upper bound

Lemma 5.1 (Uniform upper bound for the weighted input covariance). *Define the event*

$$\mathcal{E}_M := \left\{ \|M_v\|_{\text{op}} \leq \frac{6}{d} + 20p_1 \right\}.$$

Then

$$\mathbb{P}(\mathcal{E}_M) \geq 1 - e^{-d}.$$

Proof. Fix $x \in \mathbb{S}^{d-1}$. Since $\sqrt{d}x^\top v_i$ are independent standard normal variables, there are independent $\xi_i \sim \mathcal{N}(0, 1)$ such that

$$x^\top M_v x = \frac{1}{d} \sum_{i=1}^N p_i \xi_i^2.$$

Lemma 4.1 gives, for every $t > 0$,

$$x^\top M_v x \leq \frac{1}{d} \left(1 + 2 \|p\|_2 \sqrt{t} + 2p_1 t \right)$$

with failure probability at most e^{-t} .

Take a $1/4$ -net $\mathcal{N}$ of $\mathbb{S}^{d-1}$ with $|\mathcal{N}| \leq 9^d$, and choose

$$t = d(\log 9 + 1).$$

A union bound shows that the preceding estimate holds for every $x \in \mathcal{N}$ with failure probability at most

$$9^d e^{-d(\log 9 + 1)} = e^{-d}.$$

Assume this net event. Let y be a unit top eigenvector of M_v , and choose $x \in \mathcal{N}$ with $\|x - y\| \leq 1/4$. Since $M_v \succeq 0$,

$$\begin{aligned} \|M_v\|_{\text{op}} &= y^\top M_v y \\ &\leq x^\top M_v x + \left| (y - x)^\top M_v y \right| + \left| x^\top M_v (y - x) \right| \\ &\leq x^\top M_v x + \frac{1}{2} \|M_v\|_{\text{op}}. \end{aligned}$$

Thus

$$\|M_v\|_{\text{op}} \leq \frac{2}{d} \left(1 + 2 \|p\|_2 \sqrt{t} + 2p_1 t \right).$$

Because

$$\|p\|_2^2 = \sum_i p_i^2 \leq p_1 \sum_i p_i = p_1$$

and $t < 4d$,

$$\|M_v\|_{\text{op}} \leq \frac{2}{d} + 8\sqrt{\frac{p_1}{d}} + 16p_1.$$

Using $8\sqrt{ab} \leq 4a + 4b$ gives

$$\|M_v\|_{\text{op}} \leq \frac{6}{d} + 20p_1.$$

□

Proposition 5.2 (Frobenius upper estimate on explicit events). *On the event*

$$\left\{ \max_{j \in [N]} \|u_j\| \leq 2 \right\} \cap \mathcal{E}_M,$$

one has

$$\beta_F \leq 80 \left(\frac{1}{d} + p_1 \right).$$

Proof. On the displayed event, $R_u \leq 4$. Lemmas 3.2 and 5.1 give

$$\beta_F \leq 4 \left(\frac{6}{d} + 20p_1 \right) = \frac{24}{d} + 80p_1 \leq 80 \left(\frac{1}{d} + p_1 \right).$$

□

6 Two-output selectors

Lemma 6.1 (Deterministic two-output selector). *Suppose two output vectors u_a, u_b satisfy*

$$\frac{1}{2} \leq \|u_a\|^2, \|u_b\|^2 \leq \frac{3}{2}, \quad \left| u_a^\top u_b \right| \leq \frac{1}{4}.$$

Then there exists $y \in \mathbb{R}^d$ such that

$$u_a^\top y = u_b^\top y = 1, \quad \|y\|^2 \leq 8.$$

If, in addition,

$$u_j^\top y \leq \frac{1}{2} \quad \text{for every } j \notin \{a, b\},$$

then, as $t \rightarrow \infty$,

$$\text{softmax}(tU^\top y) \longrightarrow \frac{1}{2}e_a + \frac{1}{2}e_b,$$

and the associated output covariance converges to

$$C_\infty = \frac{1}{4}(u_a - u_b)(u_a - u_b)^\top.$$

For the unit vector

$$x := \frac{u_a - u_b}{\|u_a - u_b\|},$$

one has

$$x^\top C_\infty x \geq \frac{1}{8}.$$

Proof. Let

$$A := [u_a, u_b], \quad G := A^\top A.$$

By the Gershgorin bound for the symmetric 2×2 matrix G ,

$$\lambda_{\min}(G) \geq \min\{\|u_a\|^2, \|u_b\|^2\} - \left| u_a^\top u_b \right| \geq \frac{1}{4}.$$

Thus G is invertible. Define

$$y := AG^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$A^\top y = GG^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

and

$$\|y\|^2 = (1, 1)G^{-1}(1, 1)^\top \leq \frac{\|(1, 1)\|^2}{\lambda_{\min}(G)} \leq 8.$$

The selected logits are both equal to t , while every other logit is at most $t/2$. Dividing numerator and denominator of the softmax by e^t proves the softmax limit. The covariance of the limiting two-point uniform distribution is

$$\begin{aligned} C_\infty &= \frac{1}{2}u_a u_a^\top + \frac{1}{2}u_b u_b^\top - \frac{1}{4}(u_a + u_b)(u_a + u_b)^\top \\ &= \frac{1}{4}(u_a - u_b)(u_a - u_b)^\top. \end{aligned}$$

Finally,

$$\|u_a - u_b\|^2 = \|u_a\|^2 + \|u_b\|^2 - 2u_a^\top u_b \geq \frac{1}{2} + \frac{1}{2} - 2 \cdot \frac{1}{4} = \frac{1}{2},$$

so

$$x^\top C_\infty x = \frac{1}{4} \|u_a - u_b\|^2 \geq \frac{1}{8}.$$

□

Lemma 6.2 (Simultaneous selector event). *Fix $K \geq 1$ disjoint output pairs*

$$(u_1, u_2), \dots, (u_{2K-1}, u_{2K}), \quad 2K \leq N.$$

There is an event $\mathcal{E}_{\text{sel}}$ such that

$$\mathbb{P}(\mathcal{E}_{\text{sel}}) \geq 1 - \left(4Ke^{-d/32} + 2Ke^{-d/48} + K(N-2)e^{-d/64} \right),$$

and, on $\mathcal{E}_{\text{sel}}$, every pair admits vectors y_r, x_r and a covariance limit $C_{\infty, r}$ satisfying all conclusions of Lemma 6.1.

Proof. For each of the $2K$ paired vectors, the norm condition $\|u_j\|^2 \in [1/2, 3/2]$ fails with probability at most $2e^{-d/32}$. Hence all paired norm conditions hold except on an event of probability at most

$$4Ke^{-d/32}.$$

For each pair (u_{2r-1}, u_{2r}) , Lemma 4.3 gives

$$\mathbb{P} \left(\left| u_{2r-1}^\top u_{2r} \right| > \frac{1}{4}, \quad \|u_{2r-1}\|^2 \leq \frac{3}{2} \right) \leq 2e^{-d/48}.$$

Thus the probability of a pairwise inner-product failure while the relevant norm upper bound holds is at most $2Ke^{-d/48}$.

On the complement of these two failure events, Lemma 6.1 constructs y_r , depending only on (u_{2r-1}, u_{2r}) , with $\|y_r\|^2 \leq 8$. For $j \notin \{2r-1, 2r\}$, the vector u_j is independent of this pair, and therefore, conditionally on the pair,

$$u_j^\top y_r \sim \mathcal{N} \left(0, \frac{\|y_r\|^2}{d} \right).$$

The one-sided Gaussian tail bound implies

$$\mathbb{P}\left(u_j^\top y_r > \frac{1}{2} \mid u_{2r-1}, u_{2r}\right) \leq \exp\left(-\frac{(1/2)^2}{2(8/d)}\right) = e^{-d/64}.$$

Taking conditional expectations and union bounding over the $K(N-2)$ pair-outside-index choices gives the third failure term. On the resulting event, every pair satisfies the hypotheses of Lemma 6.1. $\square$

7 Frobenius lower bound

Let

$$V_K := [v_1, \dots, v_K] \in \mathbb{R}^{d \times K},$$

and define the rectangular input event

$$\mathcal{E}_V := \left\{ \sigma_{\min}(V_K) \geq \frac{1}{2}, \quad \|V_K\|_{\text{op}} \leq \frac{3}{2} \right\}.$$

By Lemma 4.4,

$$\mathbb{P}(\mathcal{E}_V) \geq 1 - 2 \cdot 17^K e^{-d/32}. \quad (2)$$

Lemma 7.1 (Single-atom Frobenius lower bound). *On $\mathcal{E}_{\text{sel}} \cap \mathcal{E}_V$,*

$$\beta_F \geq \frac{p_1}{32}.$$

Proof. Use the selector $y_1, x_1, C_{\infty,1}$ for the first output pair (u_1, u_2) . On $\mathcal{E}_V$,

$$\|v_1\| = \|V_K e_1\| \geq \sigma_{\min}(V_K) \geq \frac{1}{2},$$

so $\|v_1\|^2 \geq 1/4$. Define

$$W_t := t y_1 \frac{v_1^\top}{\|v_1\|^2}, \quad H := x_1 \frac{v_1^\top}{\|v_1\|}.$$

Then

$$W_t v_1 = t y_1, \quad \|H\|_F = 1, \quad H v_1 = x_1 \|v_1\|.$$

All Hessian summands are nonnegative. Therefore

$$\nabla^2 L(W_t)[H, H] \geq p_1 \|v_1\|^2 x_1^\top C_1(W_t) x_1.$$

As $t \rightarrow \infty$, the selector property gives $C_1(W_t) \rightarrow C_{\infty,1}$. Hence

$$\beta_F \geq p_1 \|v_1\|^2 x_1^\top C_{\infty,1} x_1 \geq p_1 \cdot \frac{1}{4} \cdot \frac{1}{8} = \frac{p_1}{32}.$$

$\square$

Lemma 7.2 (Weighted trace lower bound). *Define*

$$\mathcal{E}_{\text{tr}} := \left\{ \text{Tr}(M_v) \geq \frac{1}{2} \right\}.$$

Then

$$\mathbb{P}(\mathcal{E}_{\text{tr}}) \geq 1 - e^{-d/16}.$$

Proof. Write

$$\text{Tr}(M_v) = \frac{1}{d} \sum_{i=1}^N \sum_{k=1}^d p_i g_{ik}^2,$$

where all g_{ik} are independent standard normal variables. Apply the lower-tail part of Lemma 4.1 to the weights

$$a_{ik} := \frac{p_i}{d}.$$

Their Euclidean norm is

$$\|a\|_2 = \frac{\|p\|_2}{\sqrt{d}}.$$

Take

$$t := \frac{d}{16 \|p\|_2^2}.$$

Then $2 \|a\|_2 \sqrt{t} = 1/2$, and therefore

$$\mathbb{P} \left(\text{Tr}(M_v) < \frac{1}{2} \right) \leq \exp \left(-\frac{d}{16 \|p\|_2^2} \right) \leq e^{-d/16},$$

because $\|p\|_2^2 \leq (\sum_i p_i)^2 = 1$. □

Lemma 7.3 (Diffuse Frobenius lower bound). *On $\mathcal{E}_{\text{sel}} \cap \mathcal{E}_{\text{tr}}$,*

$$\beta_F \geq \frac{1}{32d}.$$

Proof. Use the selector $y_1, x_1, C_{\infty,1}$ for the first output pair. Gaussian input vectors are nonzero almost surely; we work on this full-probability event. For $b \in \mathbb{S}^{d-1}$, define

$$M_+(b) := \sum_{i: b^\top v_i > 0} p_i v_i v_i^\top.$$

Let b be uniform on $\mathbb{S}^{d-1}$, independently of the embeddings. For every fixed nonzero v_i , symmetry gives

$$\mathbb{P}_b(b^\top v_i > 0) = \frac{1}{2}.$$

Thus

$$\mathbb{E}_b \text{Tr}(M_+(b)) = \frac{1}{2} \text{Tr}(M_v).$$

The finite union of hyperplanes $\{b : b^\top v_i = 0\}$ has spherical measure zero. Hence there exists a b outside all these hyperplanes such that

$$\text{Tr}(M_+(b)) \geq \frac{1}{2} \text{Tr}(M_v).$$

On $\mathcal{E}_{\text{tr}}$, this gives

$$\text{Tr}(M_+(b)) \geq \frac{1}{4}.$$

Let $a \in \mathbb{S}^{d-1}$ be a top eigenvector of $M_+(b)$. Since a positive-semidefinite $d \times d$ matrix has largest eigenvalue at least its trace divided by d ,

$$a^\top M_+(b) a \geq \frac{1}{4d}.$$

Define

$$W_t := ty_1 b^\top, \quad H := x_1 a^\top.$$

Then $\|H\|_F = 1$. If $b^\top v_i > 0$, then

$$W_t v_i = t(b^\top v_i) y_1,$$

and the scalar $t(b^\top v_i)$ tends to $+\infty$. Hence

$$C_i(W_t) \longrightarrow C_{\infty,1}.$$

Using nonnegativity of all Hessian terms and the finiteness of the selected index set,

$$\begin{aligned} \beta_F &\geq \liminf_{t \rightarrow \infty} \nabla^2 L(W_t)[H, H] \\ &\geq \sum_{i: b^\top v_i > 0} p_i (a^\top v_i)^2 x_1^\top C_{\infty,1} x_1 \\ &= x_1^\top C_{\infty,1} x_1 a^\top M_+(b) a \\ &\geq \frac{1}{8} \cdot \frac{1}{4d} = \frac{1}{32d}. \end{aligned}$$

□

Proposition 7.4 (Frobenius lower estimate on explicit events). *On*

$$\mathcal{E}_{\text{sel}} \cap \mathcal{E}_V \cap \mathcal{E}_{\text{tr}},$$

one has

$$\beta_F \geq \frac{1}{64} \left(p_1 + \frac{1}{d} \right).$$

Proof. Lemmas 7.1 and 7.3 give

$$\beta_F \geq \max \left\{ \frac{p_1}{32}, \frac{1}{32d} \right\}.$$

Since $\max\{a, b\} \geq (a + b)/2$,

$$\beta_F \geq \frac{1}{64} \left(p_1 + \frac{1}{d} \right).$$

□

8 Operator-norm lower bound

Define, for $i \in [K]$,

$$g_i := \frac{u_{2i-1} - u_{2i}}{\sqrt{2}}, \quad G_K := [g_1, \dots, g_K].$$

The columns g_i are independent $\mathcal{N}(0, I_d/d)$ vectors. Define

$$\mathcal{E}_G := \left\{ \sigma_{\min}(G_K) \geq \frac{1}{2}, \quad \|G_K\|_{\text{op}} \leq \frac{3}{2} \right\}.$$

Lemma 4.4 gives

$$\mathbb{P}(\mathcal{E}_G) \geq 1 - 2 \cdot 17^K e^{-d/32}. \tag{3}$$

Proposition 8.1 (Simultaneous activation). *On*

$$\mathcal{E}_{\text{sel}} \cap \mathcal{E}_V \cap \mathcal{E}_G,$$

one has

$$\beta_{\text{op}} \geq \frac{1}{288} \sum_{i=1}^K p_i.$$

Proof. For each $i \in [K]$, let $y_i, x_i, C_{\infty, i}$ be the selector objects for the output pair (u_{2i-1}, u_{2i}) . By construction,

$$x_i = \frac{u_{2i-1} - u_{2i}}{\|u_{2i-1} - u_{2i}\|} = \frac{g_i}{\|g_i\|},$$

and

$$x_i^\top C_{\infty, i} x_i \geq \frac{1}{8}.$$

Let

$$X_K := [x_1, \dots, x_K], \quad Y_K := [y_1, \dots, y_K].$$

On $\mathcal{E}_G$, every $\|g_i\| \geq \sigma_{\min}(G_K) \geq 1/2$. Writing

$$D := \text{Diag}(\|g_1\|, \dots, \|g_K\|), \quad X_K = G_K D^{-1},$$

we obtain

$$\|X_K\|_{\text{op}} \leq \|G_K\|_{\text{op}} \|D^{-1}\|_{\text{op}} \leq \frac{3}{2} \cdot 2 = 3.$$

On $\mathcal{E}_V$, V_K has full column rank and

$$\|V_K^\dagger\|_{\text{op}} = \frac{1}{\sigma_{\min}(V_K)} \leq 2,$$

where $V_K^\dagger = (V_K^\top V_K)^{-1} V_K^\top$. Define

$$W_t := t Y_K V_K^\dagger.$$

Since $V_K^\dagger V_K = I_K$,

$$W_t v_i = t y_i \quad (i \in [K]).$$

Therefore

$$C_i(W_t) \longrightarrow C_{\infty, i} \quad (i \in [K]).$$

Now define

$$H := \frac{1}{6} X_K V_K^\dagger.$$

Then

$$\|H\|_{\text{op}} \leq \frac{1}{6} \|X_K\|_{\text{op}} \|V_K^\dagger\|_{\text{op}} \leq 1,$$

and

$$H v_i = \frac{1}{6} x_i \quad (i \in [K]).$$

Using nonnegativity of the Hessian summands,

$$\begin{aligned}
\beta_{\text{op}} &\geq \liminf_{t \rightarrow \infty} \nabla^2 L(W_t)[H, H] \\
&\geq \sum_{i=1}^K p_i \left(\frac{1}{6} x_i \right)^\top C_{\infty, i} \left(\frac{1}{6} x_i \right) \\
&\geq \frac{1}{36} \cdot \frac{1}{8} \sum_{i=1}^K p_i = \frac{1}{288} \sum_{i=1}^K p_i.
\end{aligned}$$

□

Lemma 8.2 (Head-mass comparison). *Let $a_1 \geq \dots \geq a_m \geq 0$, and let $1 \leq K \leq m$. Then*

$$\sum_{i=1}^K a_i \geq \frac{K}{m} \sum_{i=1}^m a_i.$$

Proof. If $K = m$, the claim is equality. Assume $K < m$. Since

$$a_{K+1} \leq a_K \leq \frac{1}{K} \sum_{i=1}^K a_i,$$

we have

$$\sum_{i=K+1}^m a_i \leq (m-K)a_{K+1} \leq \frac{m-K}{K} \sum_{i=1}^K a_i.$$

Adding the head mass to both sides and rearranging proves the result. □

Proposition 8.3 (Operator lower estimate on explicit events). *Assume $m \geq 512$. On*

$$\mathcal{E}_{\text{sel}} \cap \mathcal{E}_V \cap \mathcal{E}_G \cap \mathcal{E}_{\text{tr}},$$

one has

$$\beta_{\text{op}} \geq \frac{1}{294912} \left(P_m + \frac{1}{d} \right).$$

Proof. Because $m/256 \geq 2$,

$$K = \left\lfloor \frac{m}{256} \right\rfloor \geq \frac{m}{512}.$$

Apply Lemma 8.2 to the nonincreasing sequence $p_1, \dots, p_m$. Proposition 8.1 gives

$$\beta_{\text{op}} \geq \frac{1}{288} \sum_{i=1}^K p_i \geq \frac{1}{288} \frac{K}{m} P_m \geq \frac{P_m}{147456}.$$

Moreover, the Frobenius unit ball is contained in the operator-norm unit ball, because $\|H\|_{\text{op}} \leq \|H\|_F$. Hence

$$\beta_{\text{op}} \geq \beta_F.$$

On the displayed event, Lemma 7.3 therefore gives

$$\beta_{\text{op}} \geq \frac{1}{32d}.$$

Combining the two estimates,

$$\begin{aligned}\beta_{\text{op}} &\geq \max \left\{ \frac{P_m}{147456}, \frac{1}{32d} \right\} \\ &\geq \frac{P_m}{294912} + \frac{1}{64d} \\ &\geq \frac{1}{294912} \left(P_m + \frac{1}{d} \right).\end{aligned}$$

□

9 Uniform spectral-width estimate

Lemma 9.1 (Uniform block spectral-width bound). *Let $I \subseteq [N]$ be deterministic, with $s := |I| \geq 1$. Fix an output realization satisfying*

$$\max_{j \in [N]} \|u_j\| \leq 2.$$

For $\tau > 0$, define

$$T(\tau) := d^2 \log(3d) + \tau, \quad b_s(\tau) := \max \left\{ 8, \frac{2T(\tau)}{s} \right\}.$$

Conditionally on this output realization, there is an event $\mathcal{E}_I(\tau)$, depending only on $\{v_i : i \in I\}$, such that

$$\mathbb{P}(\mathcal{E}_I(\tau) \mid U) \geq 1 - e^{-\tau},$$

and, on

$$\mathcal{E}_I(\tau) \cap \left\{ \max_{i \in I} \|v_i\| \leq 2 \right\},$$

one has

$$\sup_{\|H\|_{\text{op}} \leq 1} \frac{1}{s} \sum_{i \in I} w_U(Hv_i)^2 \leq \frac{64}{d} (L_N + b_s(\tau) + 2).$$

Proof. Fix H with $\|H\|_{\text{op}} \leq 1$, and let $v \sim \mathcal{N}(0, I_d/d)$, independently of U . For fixed j, k , conditionally on U ,

$$(u_j - u_k)^\top H v \sim \mathcal{N} \left(0, \frac{\|H^\top(u_j - u_k)\|^2}{d} \right).$$

Because $\|u_j - u_k\| \leq 4$, the conditional variance is at most $16/d$. Therefore

$$\mathbb{P} \left(\left| (u_j - u_k)^\top H v \right| \geq z \mid U \right) \leq 2e^{-dz^2/32}.$$

A union bound over at most N^2 ordered pairs gives

$$\mathbb{P}(w_U(Hv) \geq z \mid U) \leq 2N^2 e^{-dz^2/32}.$$

With

$$z^2 = \frac{32}{d}(L_N + r), \quad L_N = \log(2N^2),$$

we obtain

$$\mathbb{P} \left(w_U(Hv)^2 \geq \frac{32}{d}(L_N + r) \mid U \right) \leq e^{-r}.$$

For $i \in I$, define

$$Y_i := \left(\frac{d}{32} w_U(Hv_i)^2 - L_N \right)_+.$$

Then, conditionally on U , the variables Y_i are independent and

$$\mathbb{P}(Y_i \geq r \mid U) \leq e^{-r} \quad (r \geq 0).$$

For $0 < \lambda < 1$, the tail-integration identity gives

$$\begin{aligned} \mathbb{E}[e^{\lambda Y_i} \mid U] &= 1 + \int_0^\infty \lambda e^{\lambda r} \mathbb{P}(Y_i \geq r \mid U) dr \\ &\leq 1 + \lambda \int_0^\infty e^{-(1-\lambda)r} dr = \frac{1}{1-\lambda}. \end{aligned}$$

Thus, for $b > 1$, Chernoff's inequality with $\lambda = 1 - 1/b$ yields

$$\mathbb{P}\left(\frac{1}{s} \sum_{i \in I} Y_i \geq b \mid U\right) \leq e^{-s(b-1-\log b)}.$$

For $b \geq 8$,

$$b - 1 - \log b \geq \frac{b}{2}.$$

We claim that

$$s(b_s(\tau) - 1 - \log b_s(\tau)) \geq T(\tau).$$

Indeed, if $b_s(\tau) = 2T(\tau)/s$, the claim follows from the preceding half-linear bound. If $b_s(\tau) = 8$, then $T(\tau) \leq 4s$, while $8 - 1 - \log 8 > 4$. This proves the claim.

Take a $d^{-1/2}$ -net $\mathcal{H}$ of the operator unit ball. By Lemma 4.2,

$$|\mathcal{H}| \leq (3d)^{d^2}.$$

A union bound over $H_0 \in \mathcal{H}$ shows that, with conditional failure probability at most

$$(3d)^{d^2} e^{-T(\tau)} = e^{-\tau},$$

one has simultaneously

$$\frac{1}{s} \sum_{i \in I} w_U(H_0 v_i)^2 \leq \frac{32}{d} (L_N + b_s(\tau)) \quad (H_0 \in \mathcal{H}).$$

This defines $\mathcal{E}_I(\tau)$.

Now let $\|H\|_{\text{op}} \leq 1$, and choose $H_0 \in \mathcal{H}$ satisfying $\|H - H_0\|_{\text{op}} \leq d^{-1/2}$. On the input-norm event in the statement,

$$\begin{aligned} w_U(Hv_i) &\leq w_U(H_0 v_i) + \max_{j,k} \left| (u_j - u_k)^\top (H - H_0) v_i \right| \\ &\leq w_U(H_0 v_i) + 4d^{-1/2} \|v_i\| \\ &\leq w_U(H_0 v_i) + \frac{8}{\sqrt{d}}. \end{aligned}$$

Therefore

$$w_U(Hv_i)^2 \leq 2w_U(H_0v_i)^2 + \frac{128}{d}.$$

Average over $i \in I$ and use the net estimate to obtain

$$\frac{1}{s} \sum_{i \in I} w_U(Hv_i)^2 \leq \frac{64}{d} (L_N + b_s(\tau) + 2).$$

Taking the supremum over H proves the result. $\square$

10 Operator-norm upper bound

Define the global norm event

$$\mathcal{E}_{\text{norm}} := \left\{ \max_{j \in [N]} \|u_j\| \leq 2, \quad \max_{i \in [N]} \|v_i\| \leq 2 \right\}.$$

By Lemma 4.3 and a union bound,

$$\mathbb{P}(\mathcal{E}_{\text{norm}}) \geq 1 - 2Ne^{-d/2}. \quad (4)$$

Assume $N > d$. For $r = 0, \dots, J-1$, set

$$a_r := 2^r d, \quad I_r := \{a_r + 1, \dots, \min(2a_r, N)\}, \quad s_r := |I_r|.$$

These nonempty, disjoint blocks partition $\{d+1, \dots, N\}$. For each block, apply Lemma 9.1 with $\tau = d$. Since $T(d) = T_d$, conditionally on any output realization satisfying the output part of $\mathcal{E}_{\text{norm}}$, the simultaneous block event

$$\mathcal{E}_{\text{width}} := \bigcap_{r=0}^{J-1} \mathcal{E}_{I_r}(d)$$

satisfies

$$\mathbb{P}(\mathcal{E}_{\text{width}}^c \mid U) \leq Je^{-d}. \quad (5)$$

When $N \leq d$, define $\mathcal{E}_{\text{width}}$ to be the whole sample space.

Proposition 10.1 (Operator upper estimate on explicit events). *On $\mathcal{E}_{\text{norm}}$,*

$$\beta_{\text{op}} \leq 16.$$

If $N > d$, then on $\mathcal{E}_{\text{norm}} \cap \mathcal{E}_{\text{width}}$,

$$\beta_{\text{op}} \leq 16P_d + \frac{2^{\alpha+4}(L_N + 10)}{d} + \frac{32T_d}{d^2} \Xi_{\alpha, J} P_d.$$

Proof. On $\mathcal{E}_{\text{norm}}$,

$$R_u \leq 4, \quad \text{Tr}(M_v) = \sum_{i=1}^N p_i \|v_i\|^2 \leq 4.$$

The second estimate in Lemma 3.3 gives

$$\beta_{\text{op}} \leq 16.$$

Assume now that $N > d$. For $i \leq d$ and $\|H\|_{\text{op}} \leq 1$,

$$w_U(Hv_i) \leq \max_{j,k} \|u_j - u_k\| \|Hv_i\| \leq 4 \cdot 2 = 8.$$

Thus the head contribution in Lemma 3.3 is at most

$$\frac{1}{4} \sum_{i=1}^d p_i w_U(Hv_i)^2 \leq 16P_d.$$

Fix a tail block I_r . Since p_i is nonincreasing,

$$\sum_{i \in I_r} p_i w_U(Hv_i)^2 \leq p_{a_r+1} \sum_{i \in I_r} w_U(Hv_i)^2.$$

The block-width estimate with $\tau = d$ yields

$$\begin{aligned} \frac{1}{4} \sum_{i \in I_r} p_i w_U(Hv_i)^2 &\leq \frac{16}{d} p_{a_r+1} s_r \left[L_N + \max \left\{ 8, \frac{2T_d}{s_r} \right\} + 2 \right] \\ &\leq \frac{16(L_N + 10)}{d} s_r p_{a_r+1} + \frac{32T_d}{d} p_{a_r+1}. \end{aligned}$$

For every $i \in I_r$, we have $i \leq 2a_r$, and therefore

$$p_i = \frac{i^{-\alpha}}{H_{N,\alpha}} \geq 2^{-\alpha} \frac{a_r^{-\alpha}}{H_{N,\alpha}} \geq 2^{-\alpha} p_{a_r+1}.$$

Hence

$$s_r p_{a_r+1} \leq 2^\alpha \sum_{i \in I_r} p_i,$$

and summing over blocks gives

$$\sum_{r=0}^{J-1} s_r p_{a_r+1} \leq 2^\alpha.$$

Also,

$$p_{a_r+1} \leq \frac{a_r^{-\alpha}}{H_{N,\alpha}} = p_d 2^{-\alpha r}.$$

Since $P_d \geq dp_d$,

$$\sum_{r=0}^{J-1} p_{a_r+1} \leq p_d \Xi_{\alpha,J} \leq \frac{P_d}{d} \Xi_{\alpha,J}.$$

Summing the preceding block estimate over r gives

$$\frac{1}{4} \sup_{\|H\|_{\text{op}} \leq 1} \sum_{i=d+1}^N p_i w_U(Hv_i)^2 \leq \frac{2^{\alpha+4}(L_N + 10)}{d} + \frac{32T_d}{d^2} \Xi_{\alpha,J} P_d.$$

Adding the head contribution and invoking Lemma 3.3 proves the claim. $\square$

11 Proof of the main theorem

Proof of Theorem 1.1. Because $m \geq 512$, we have $K \geq 1$, $K \leq d/256$, and $2K \leq N$. Thus all selector and rectangular-matrix constructions above are valid.

Consider the event

$$\mathcal{G} := \mathcal{E}_{\text{norm}} \cap \mathcal{E}_M \cap \mathcal{E}_{\text{tr}} \cap \mathcal{E}_{\text{sel}} \cap \mathcal{E}_V \cap \mathcal{E}_G \cap \mathcal{E}_{\text{width}}.$$

The individual failure bounds are:

$$\begin{aligned} \mathbb{P}(\mathcal{E}_{\text{norm}}^c) &\leq 2Ne^{-d/2}, \\ \mathbb{P}(\mathcal{E}_M^c) &\leq e^{-d}, \\ \mathbb{P}(\mathcal{E}_{\text{tr}}^c) &\leq e^{-d/16}, \\ \mathbb{P}(\mathcal{E}_{\text{sel}}^c) &\leq 4Ke^{-d/32} + 2Ke^{-d/48} + K(N-2)e^{-d/64}, \\ \mathbb{P}(\mathcal{E}_V^c) + \mathbb{P}(\mathcal{E}_G^c) &\leq 4 \cdot 17^K e^{-d/32}. \end{aligned}$$

For the width event, let

$$\mathcal{E}_{\text{norm}}^U := \left\{ \max_j \|u_j\| \leq 2 \right\}.$$

Equation (5) and the tower property imply

$$\mathbb{P}(\mathcal{E}_{\text{norm}}^U \cap \mathcal{E}_{\text{width}}^c) \leq Je^{-d}.$$

Therefore a union bound gives

$$\mathbb{P}(\mathcal{G}^c) \leq \varepsilon_{\text{tot}}(d, N),$$

with ε_{tot} defined in (1).

On $\mathcal{G}$, Proposition 5.2 gives the Frobenius upper bound, and Proposition 7.4 gives the Frobenius lower bound. Proposition 8.3 gives the operator lower bound. Proposition 10.1 gives both the universal operator upper bound 16 and, when $N > d$, the sharper dyadic upper bound. Hence all conclusions of Theorem 1.1 hold simultaneously on $\mathcal{G}$. $\square$

12 Zipf consequences in the polynomial-size regime

This section records consequences of Theorem 1.1; it is not used in its proof.

Corollary 12.1 (Polynomial-size summary). *Fix $\alpha \geq 0$. Suppose $N = d^{\Theta(1)}$ and $m = \min\{d, N\} \geq 512$. Then*

$$\varepsilon_{\text{tot}}(d, N) = e^{-\Omega(d)}.$$

On the event in Theorem 1.1,

$$\beta_F = \Theta\left(\frac{1}{d} + p_1\right),$$

with universal implicit constants, and

$$\beta_{\text{op}} = \tilde{\Theta}_\alpha\left(\frac{1}{d} + P_m\right),$$

where the tilde hides factors polylogarithmic in d .

Proof. Since N is polynomial in d , all polynomial prefactors in (1) are dominated by the displayed exponential tails. For the rectangular terms, use $K \leq d/256$:

$$17^K e^{-d/32} \leq \exp \left[- \left(\frac{1}{32} - \frac{\log 17}{256} \right) d \right] = e^{-\Omega(d)}.$$

Thus $\varepsilon_{\text{tot}}(d, N) = e^{-\Omega(d)}$.

The Frobenius assertion is exactly Theorem 1.1. For the operator upper bound, if $N \leq d$, then $P_m = P_N = 1$, and the lower and upper bounds are both of constant order. If $N > d$, then $m = d$,

$$L_N = O(\log d), \quad \frac{T_d}{d^2} = O(\log d), \quad J = O(\log d),$$

and

$$\Xi_{\alpha, J} = \begin{cases} O(\log d), & \alpha = 0, \\ O_\alpha(1), & \alpha > 0. \end{cases}$$

Therefore

$$\beta_{\text{op}} \lesssim_\alpha \begin{cases} P_d \log^2 d + \frac{\log d}{d}, & \alpha = 0, \\ P_d \log d + \frac{\log d}{d}, & \alpha > 0. \end{cases}$$

Combining with the lower bound proves the stated polylogarithmic equivalence. $\square$

Corollary 12.2 (Evaluation of the Zipf head masses). *For fixed $\alpha \geq 0$,*

$$H_{n, \alpha} = \begin{cases} \Theta_\alpha(n^{1-\alpha}), & 0 \leq \alpha < 1, \\ \Theta(\log n), & \alpha = 1, \\ \Theta_\alpha(1), & \alpha > 1. \end{cases}$$

Consequently,

$$\beta_F = \begin{cases} \Theta_\alpha(d^{-1} + N^{\alpha-1}), & 0 \leq \alpha < 1, \\ \Theta(d^{-1} + (\log N)^{-1}), & \alpha = 1, \\ \Theta_\alpha(1), & \alpha > 1, \end{cases}$$

and, in the polynomial-size regime,

$$\beta_{\text{op}} = \begin{cases} \tilde{\Theta}_\alpha \left(d^{-1} + \left(\frac{m}{N} \right)^{1-\alpha} \right), & 0 \leq \alpha < 1, \\ \tilde{\Theta} \left(d^{-1} + \frac{\log m}{\log N} \right), & \alpha = 1, \\ \Theta_\alpha(1), & \alpha > 1, \end{cases} \quad m = \min\{d, N\}.$$

For $0 \leq \alpha < 1$ and $N > d$, the exact identity is

$$P_d = H_{d, \alpha} p_1,$$

and therefore

$$P_d \asymp_\alpha \left(\frac{d}{N} \right)^{1-\alpha} \asymp_\alpha d^{1-\alpha} p_1.$$

The last two relations are asymptotic comparisons, not exact equalities in general.

Proof. The harmonic-number estimates follow from elementary integral comparison. Substitute

$$p_1 = H_{N,\alpha}^{-1}, \quad P_m = \frac{H_{m,\alpha}}{H_{N,\alpha}}$$

into Theorem 1.1 and Corollary 12.1. The identity $P_d = H_{d,\alpha}p_1$ follows directly from the definitions. $\square$